Master in Digital Marketing and Communication

Course: Neurosciences Applied to Marketing

Three takeaways through the case of VR gaming

Alberto Faggiotto

Student n°0900076836

Academic year 2025/2026

INDEX

1. **Introduction** (p.2-4)

    1.1 Before *Batman*: the VR adoption problem (p.4)

2. **My three takeaways** (p.4-10)

    2.1 First takeaway: experiential marketing (p.4-6)

    2.2 Second takeaway: attention, emotion and memory (p.7-8)

    2.3 Third takeaway: reducing cognitive risk (p.9-10)

3. **Conclusions** (p.10-11)

4. **References** (p.12-13)

    4.1 Academic sources (p.12-13)

    4.2 Non-academic sources (p.13)

1. Introduction

I came to this course with a background in semiotics and one of the things that struck me most during the lectures was how familiar the underlying questions felt, even dressed differently: the “Semiotics of consumption”, which I studied at the University of Bologna, had already taught me to look at consumer behaviour as a system of meanings but what this course added was the layer of the neurological and psychological mechanics beneath those meanings. I found that combination genuinely engaging, because it sits in a productive space between scientific theory and practical application. Before starting the master I had read Goleman’s *Emotional Intelligence*, partly out of personal curiosity and partly because I suspected it would matter professionally, and the course confirmed the intuition that the emotional dimension of how people process stimuli, form preferences and make decisions is, in many ways, the core of marketing thinking, and knowing this changes how I will approach the discipline going forward.

There is also a more personal thread running through my choice of case study. I grew up playing video games and I was always drawn to the moments when the industry introduced something that changed the physical relationship between player and screen. For example, I used the Kinect on the Xbox 360 as a kid (a motion-sensing accessory for the console that allowed players to control games using body movements and voice commands, without any physical controller) and I bought PlayStation VR at launch. What interested me then was the specific quality of desire those products generated - how you could want something intensely before you had any clear sense of what it actually felt like to use it - and as an adult studying marketing that memory takes on a different meaning. The question of how a technology that most people have never tried becomes something they feel they need is a really interesting marketing problem, and video games - the sector which remains the most economically significant in the entertainment industry - have been posing it repeatedly for decades.

Fig. 1: the Kinect for Xbox 360

Fig. 2: the first VR headset for PlayStation, released in 2016

The release of *Batman: Arkham Shadow* in October 2024 felt like a good case to work through these questions. Developed by Camouflaj exclusively for Meta Quest 3 and 3S, the game was positioned by Meta as a flagship experience for the platform - a launch designed to make VR feel more accessible and more culturally legitimate to audiences sitting on the fence about the technology. Between May and October 2024, the launch unfolded in distinct steps: a first reveal in May, a story trailer at Summer Game Fest in June, a gameplay reveal at Gamescom in August, integration into the Quest 3S hardware bundle in September, the game release in October, and the physical demo tour shortly after. As Kostyk and Sheng (2022) argue, VR marketing efforts should be matched to "the objectives associated with each of the six stages of the customer journey" (p. 226) and this sequence maps fairly onto that logic, with each touchpoint serving a different function at a different moment in the consumer's relationship with the product.

Fig.3: The first promotional image for *Batman: Arkham Shadow*, in which the iconic Batman logo is rendered through the gameplay components of the new VR technology

This essay uses that launch as a concrete case organised around three takeaways from the course: the role of direct embodied experience as a marketing tool, explored through the *Arkham Shadow Immersive Demo Experience tour*; the logic of the campaign *Become the Knight* read through the attention economy and the neuroscience of emotional appeal; and the strategic function of a high-recognition franchise in reducing the cognitive and symbolic risk of committing to an unfamiliar platform. Since no neurophysiological data on the tour or the campaign exists in the public domain, the analytical framework draws on consumer neuroscience and behavioural decision research as interpretive lenses, to understand what the launch was trying to do at the level of attention, memory and affective response. But before getting to the launch, it is worth asking why a launch of this scale was necessary in the first place.

1.1 Before *Batman*: the VR adoption problem

Virtual reality has been commercially available to consumers for about a decade, and yet it has remained, for most of that time, a technology associated with a specific kind of user. Cummings et al. (2023) describe consumer-level VR as sitting in "the earliest of stages of the diffusion of innovations process" (p. 1357), with adoption driven primarily by psychological traits like immersive tendencies, sensation seeking and absorption - in other words, people already predisposed to this kind of experience. This is a significant constraint for any company trying to grow a platform, because if the people most likely to buy a VR headset are already those who would have sought it out anyway, the market has a ceiling that is hard to push against through conventional marketing.

The research on what drives adoption beyond that early-adopter group points to a specific set of variables. Lee et al. (2019) found that, among the elements of the Technology Acceptance Model applied to VR, "perceived enjoyment had the largest total effect on the intention to use" (p. 46), and more than perceived usefulness or ease of use; this shifts the marketing problem away from communicating features and toward creating conditions for pleasurable, emotionally engaging first contact with the technology. The same study also found that social interactions and the strength of social ties played a key role in amplifying that enjoyment, leading the authors to suggest that "invigorating social networks when marketing VR products and services could play a crucial role in helping them cross Moore's chasm" (*ibidem*). VR, in other words, spreads more easily when people can share the experience and talk about it.

For VR games specifically, the picture is slightly more nuanced. Kosa et al. (2020) showed that what predicts acceptance is a chain that starts with ease of use, moves through autonomy and competence satisfaction, and arrives at enjoyment, with this last one predicting both positive attitudes toward VR gaming and intention to play. The implication is that "content of VR games and user perception plays a major role in its acceptance" (Kosa et al., 2020, p. 58), beyond hardware considerations alone. A well-designed game, easy to enter and satisfying to play, does something that no amount of advertising can replicate and this is the context in which the *Batman: Arkham Shadow* launch needs to be read. Meta's challenge in late 2024 wasn't simply to release a new title but it was to make VR feel approachable, enjoyable and socially legible to people who had never seriously considered buying a headset.

2. My three takeaways

2.1 First takeaway: experiential marketing

As we know, the idea that marketing should create experiences rather than just communicate messages has been in the discipline for almost thirty years. Already in 1998 Pine and

Gilmore (*Welcome to the experience economy*) began to point out how economic value was shifting from goods and services toward experiences and that what consumers increasingly sought wasn't a product but a memorable event involving them personally. Schmitt (1999) approached the same shift from a marketing perspective, distinguishing experiential marketing from traditional feature-and-benefit approaches on the basis that it targets "sensory, emotional, cognitive, behavioural, and relational values' in the consumer rather than functional attributes of the product" (p.57) in the consumer rather than functional attributes of the product. Following this path, direct experience encodes differently from vicarious exposure since experiential marketing activates perception, attention and affective response simultaneously: the resulting memory trace is richer and more durable.

And yet, for most products, experiential marketing is a premium option, powerful but not strictly necessary, but for VR I would argue it's something closer to a structural requirement. The gap between imagining a VR experience and actually having one is too wide to bridge through conventional advertising, however well made. This is where the *Batman: Arkham Shadow Immersive Demo Experience* tour becomes a truly interesting case, because held across multiple cities in the United States and Canada in late 2024, it placed branded physical environments in accessible locations where people could put on a Quest 3S headset to play a segment of the game, and in this way the physical space was the access point to a digital experience that would otherwise have remained abstract.

Fig.4: a section of the set from the *Batman: Arkham Shadow Immersive Demo Experience* in Toronto

Zhang et al. (2025) show that immersive VR enhances purchase intention through the mediating role of *product diagnosticity*, which they define as consumers' capability to evaluate a product through the experience itself. As the authors explain, "immersive VR, with its high-quality 360-degree true-to-scale product representation, closely aligns with consumers' real-life experiences and enables detailed and thorough product inspection and

information acquisition" (p. 352), and this enhanced diagnosticity "alleviates consumers' unfamiliarity and uncertainty, fostering a more concrete mental representation of the product" (*ibidem*). For a technology where the barrier to adoption is largely imaginative and less rational, this mechanism is critical: the richer and more embodied the experience, the more concrete the consumer's mental representation of the product, and the lower the perceived risk of commitment.

Moreover, Altschwager et al. (2024) show that sensorial and pragmatic event experiences are particularly effective at generating engagement that transfers from the event to the brand, since "pragmatic event experiences specifically initiate behavioural activities that require physical immersion and interaction as the core activity" (p. 35), and sensorial ones elicit "feelings of consumer excitement, enthusiasm, positive aesthetic response, pleasure, and happiness" (*ibidem*) that, when tied to the brand, build durable engagement. Crucially, the authors find that this transfer doesn't happen automatically, since pragmatic experience can actually have a negative direct effect on brand engagement if the physical activity absorbs the participant to the point of disconnecting them from brand associations. The Arkham Shadow tour was designed to avoid exactly this risk - the thematized environment, the Batman branding and the *Become the knight* concept ensured that the physical experience of wearing the headset remained semantically inseparable from the brand proposition.

Fig. 5: another example of the immersive set, which can be seen in action HERE

The takeaway I carry from this is that experiential marketing is about identifying the moments in a consumer journey where no message can substitute for direct contact with the product and designing the experience that makes that contact possible, legible and affectively resonant.

2.2 Second takeaway: attention, emotion and memory

One of the most consistent insights to emerge from consumer neuroscience in recent years is that the brain doesn't allocate attention neutrally, but instead it works through an active and selective process governed by emotional relevance, novelty and personal significance: in an environment of chronic informational overload the ability to design communication that genuinely captures and holds attention has become one of the central problems of the discipline. In this framework, the concept of the *attention economy* is very important, since it refers to the condition in which human attention is the scarce resource and communication strategies must compete for it with increasing sophistication.

What the *Batman: Arkham Shadow* campaign did well, and what makes it interesting from this perspective, is that it didn't try to win attention through volume or frequency alone: if we consider the creative concept *Become the Knight* we can understand how it offers the consumer something worth imagining; it doesn't highlight the quality of the specification of the headset or the greatness of the game and it doesn't underlines how you will watch or control Bruce Wayne, but instead it highlights you will *be* him, working through a high-level mental representation appeal. Wu et al. (2025) show that VR advertising is significantly more effective when it uses emotional rather than rational appeals, precisely because "VR ads enable consumers to experience the psychological, social, or symbolic value of products in a virtual environment" (p. 10): *Become the Knight* is a textbook emotional appeal of exactly this kind, since it targets identity and not information.

Fig.6: The promotional image for the *Become the Knight* campaign made the complete fusion between player and masked hero immediately legible

Uysal et al. (2025) demonstrate that VR product presentations work partly through a mechanism of psychological ownership, defined as "the feeling that a product is *mine* even without actual ownership" (p. 5), and show that this sense of possession is amplified by immersion and interactivity: "interactivity with a product in a VR environment enhances the

effect of VR product presentation on psychological ownership" (p.6), as direct engagement with the product deepens the consumer's sense of connection and simulated possession before any purchase decision is made. So, from a neuroscientific standpoint, the mechanism behind *Become the Knight* matters because emotional and identity-relevant stimuli are processed with greater depth and encoded more durably in memory: the brain's attentional system prioritises stimuli that carry personal significance, and the prospect of inhabiting a culturally iconic character activates precisely this kind of self-relevant processing. *Become the Knight* was designed to trigger this anticipatory simulation at a distance, through advertising, before the headset was ever put on.

The media mix confirms that this was a deliberate attention strategy because each channel was chosen for a different attentional context, from passive lean-back environments like cinema and podcast to high-frequency touchpoints like social and DOOH: CTV, online video, paid social, OOH, podcast placements and cinema advertising. The logic was reach, of course, but also to transfer an experiential concept across contexts where people's attentional states are very different, ensuring that the idea of embodiment registered across the full audience spectrum, well beyond the existing VR community.

Fig.7: An example of OOH advertising from the *Become the Knight* campaign on the streets of Toronto

The takeaway I draw from this is that understanding how attention works neurologically changes how you evaluate a campaign: *Become the Knight* is a good piece of communication because it's structured around the mechanisms through which the brain actually encodes brand-related experience, such as emotional salience, identity relevance, and the anticipatory simulation of a desired future state.

2.3 Third takeaway: reducing cognitive risk

Deciding whether to adopt an unfamiliar technology is, cognitively speaking, a demanding task, because it requires assembling information from incomplete sources, tolerating uncertainty about outcomes, and constructing a mental model of something one has never directly experienced. This is a System 2 operation - in Kahneman's dual-process framework (*Thinking, Fast and Slow*), the slow, deliberate, effortful mode of thinking that we engage when we cannot rely on automatic pattern recognition. The problem, from a marketing standpoint, is that System 2 processing is costly, and perceived risk and cognitive effort together form a barrier to adoption that rational arguments and feature lists do little to dissolve (precisely because those arguments require more System 2 engagement to evaluate).

This is where franchise recognition enters as a strategic variable, since familiar brands and culturally embedded narratives are processed through a fundamentally different cognitive pathway: consumer neuroscience has provided converging evidence that brand familiarity significantly alters the neural architecture of preference formation. Solnais et al. (2013), reviewing the neuroimaging literature on brand familiarity, note that "the activation of the ventromedial prefrontal cortex was found to increase prior to the subject's expressed choice in the case of novel brands versus familiar brands" (p. 75), meaning that unfamiliar brands require more active deliberative processing while familiar ones engage a more automatic, affectively-driven evaluation. The same study notes that "the formation of consumer preferences doesn't solely rely on a rational evaluation of product attributes but also involves non-rational and emotional decision-making processes" (p. 75) adding that the cingulate cortex plays a role in integrating implicit emotional information about known brands into the decision process. In System 1/System 2 terms: a familiar brand partially bypasses the effortful deliberation that an unknown brand would trigger, activating instead a faster, affectively mediated response.

Applied to the *Batman: Arkham Shadow* launch, the implication is that Meta was asking consumers to evaluate a technology (VR) that most of them had limited direct experience with, in a context of non-trivial financial commitment: we can understand that is a high cognitive-cost decision and Batman reduced that cost not because it provided new information about the Quest 3S, but because it provided a pre-built affective scaffold: a narrative world already populated with emotional associations, visual familiarity and personal significance for millions of people. The consumer doesn't need to construct a mental model of the experience from scratch because they already have one, anchored in decades of cultural exposure to the character. Batman effectively recruited System 1 in a decision context that would otherwise have demanded System 2.

Fig.8: for the first time, the new VR technology has proved to be a success, establishing *Batman: Arkham Shadow* as a marketing case study worth keeping as a model for future launches.

The game's recognition as Best VR/AR Game at The Game Awards 2024 reinforced this legitimation function, providing an external validation signal that further reduced the uncertainty of committing to the platform; what we can say, with reasonable confidence, is that the franchise served as a cognitive and affective anchor that made the adoption decision easier to approach, and that this mechanism relies on how the brain actually processes brand familiarity under conditions of uncertainty.

3. Conclusions

The answer to the question of why Meta chose Batman to market virtual reality and what that choice can tell us about how immersive experiences function as marketing tools reflects my three takeaways of this class: the demo tour, the campaign and the franchise choice all work toward the goal of making an unfamiliar technology feel approachable, imaginable and worth committing to. Each layer addresses a different barrier to adoption: the tour does it through direct embodied experience, which no amount of paid media can replicate for a product whose value is mainly sensory, while the campaign does it by constructing an identity-relevant emotional appeal that activates anticipatory simulation before the headset is ever put on. The Batman franchise, in turn, does it by providing a pre-built affective scaffold that lowers the cognitive cost of evaluating something new, recruiting System 1 in a decision context that would otherwise require System 2.

My background had always been primarily interpretive (semiotics, cultural criticism, close reading of texts and images) and my previous tools were good at explaining what a communication means, how it positions the consumer or what symbolic world it constructs; however, what they tend to skip is the unconscious layer, that is to say what happens before interpretation, in the automatic and affective responses that precede conscious meaning-making; that is exactly where neuroscience operates, and finding that the two perspectives are complementary was, for me, the most intellectually satisfying part of the course. In fact, more than anything, this course pushed the interpretive thinking I already had

toward a more operational level: this shift matters particularly for the professional direction I want to pursue, since I'm interested in working in entertainment communication (film, television, video games) and writing this essay made clearer to me how central neuroscientific thinking is to building successful products in that space. The questions that the industry is actually trying to answer - why certain franchises generate desire before anyone has seen a frame of footage, how immersive experiences convert curiosity into commitment, what makes a campaign land emotionally - are neuroscience questions as much as they are creative ones, and this is why I think I have obtained even more solid foundations to build my future.

4. References

4.1 Academic sources

Altschwager, T., Conduit, J., Karpen, I. O., & Goodman, S. (2024). Event engagement: Using event experiences to build brands. *Australasian Marketing Journal, 32*(1), 31–44. https://doi.org/10.1177/14413582221127317

Cummings, J. J., Cahill, T. J., Wertz, E., & Zhong, Q. (2023). Psychological predictors of consumer-level virtual reality technology adoption and usage. *Virtual Reality, 27*, 1357–1379. https://doi.org/10.1007/s10055-022-00736-1

Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux

Kosa, M., Uysal, A., & Eren, P. E. (2020). Acceptance of virtual reality games: A multi-theory approach. *International Journal of Gaming and Computer-Mediated Simulations, 12*(1), 43–60. https://doi.org/10.4018/IJGCMS.2020010103

Kostyk, A., & Sheng, J. (2022). VR in customer-centered marketing. *Business Horizons, 65*(2), 223–234. https://doi.org/10.1016/j.bushor.2021.02.027

Lee, J., Kim, J., & Choi, J. Y. (2019). The adoption of virtual reality devices: The technology acceptance model integrating enjoyment, social interaction, and strength of the social ties. *Telematics and Informatics, 39*, 37–48. https://doi.org/10.1016/j.tele.2018.12.006

Pine, J., & Gilmore, J. H. (1998). Welcome to the experience economy. *Harvard Business Review, 76*(4), 97–105.

Schmitt, B. (1999). Experiential marketing. *Journal of Marketing Management, 15*(1–3), 53–67. https://doi.org/10.1362/026725799784870496

Solnais, C., Andreu-Perez, J., Sánchez-Fernández, J., & Andréu-Abela, J. (2013). The contribution of neuroscience to consumer research: A conceptual framework and empirical review. *Journal of Economic Psychology, 36*, 68–81. https://doi.org/10.1016/j.joep.2013.02.011

Uysal, E., Finken, D., Krämer, M., Alavi, S., Jun, Y., & Schendzielarz, D. (2025). Virtually mine: Understanding consumer responses to virtual reality product presentations. *Journal of Retailing*. https://doi.org/10.1016/j.jretai.2025.04.007

Wu, J.-F., Zhang, Z., Wu, Y., & Tang, X.-J. (2025). The effectiveness of immersive media in promoting consumer products: Augmented vs. virtual reality. *Journal of Retailing and Consumer Services, 86*, 104323. https://doi.org/10.1016/j.jretconser.2025.104323

Zhang, Y., Shao, W., Quach, S., Thaichon, P., & Li, Q. (2025). VR retailing: When and why immersion pays off. *Journal of Retailing, 101*, 348–365. https://doi.org/10.1016/j.jretai.2025.04.004

4.2 Non-academic sources

Camouflaj. (2024, August). *Batman: Arkham Shadow — gameplay reveal, Gamescom 2024* [Video]. YouTube.

MediaPost / Muse by Clio. (2024). *"Become the Knight" — Batman: Arkham Shadow campaign coverage*.

Meta Platforms. (2024, May). *Batman: Arkham Shadow — official reveal*. Meta.com.

Meta Platforms. (2024, September). *Meta Quest 3S launch and Batman: Arkham Shadow bundle*. Meta.com.

The Game Awards. (2024). *Best VR/AR Game: Batman: Arkham Shadow*. thegameawards.com.

UploadVR. (2024, November). *Batman: Arkham Shadow Immersive Demo Experience tour*. UploadVR.com.